Taewoon Kang

Integrated M.S. and Ph.D. Student
Korea University

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Education

Korea University Mar. 2023 – Present
Department of Computer Science and Engineering Seoul, Korea
Ph.D. Student | GPA: 4.35 / 4.5
Advisor: Gunjae Koo

Sangmyung University Mar. 2019 – Aug. 2022
Department of System Semiconductor Engineering Cheonan, Korea
B.S. in System Semiconductor Engineering | GPA: 4.28 / 4.5
Advisor: Yongwoo Kim
Thesis: *An Implementation of CNN-based Real-Time Face Mask Detector using Jetson Nano*

Research Interests

AI accelerators; HW/SW co-design for SoCs; processing for AI (NPU); computer architecture; processing-in-memory.

Experience

University of Southern California Jan. 2026 – Present
Visiting Scholar United States
Ming Hsieh Department of Electrical and Computer Engineering
Supervisor: Prof. Murali Annavam

SK Telecom Jun. 2023 – Aug. 2023
Intern Korea
Developed spam detection methods using AI.

Teaching Experience

Teaching Assistant

Basic of software Spring 2025
Digital logic design Spring 2023, Fall 2023

Academic Service

Junior Program Committee 2027
ACM International Conference on Supercomputing (ICS)

Artifact Evaluation Committee 2026
IEEE/ACM International Symposium on Microarchitecture (MICRO)

Publications | International

1. Narae Kim, Wonjae Choi, **Taewoon Kang**, Yujin Kim, Seongpil Yang, Myung Kuk Yoon, Yunho Oh, Gunjae Koo. "Fine-Grained DRAM Refresh Exploiting Divergent Bit-Error Tolerance in AI Workloads." [Under review]
2. **Taewoon Kang**, Jonghyun Lee, Yujin Kim, Murali Annavaram, Gunjae Koo. "LogPIM: Low-Overhead Logarithmic Arithmetic for HBM-Based Processing-in-Memory." *IEEE Computer Architecture Letters*. [To appear]
3. Jong Hyun Jeong, Yuan Feng, **Taewoon Kang**, Gunjae Koo, Hyeran Jeon. "GPO: Orchestrating GPU-PIM Computing for Sparse-Dense Workload Acceleration." [Under review]
4. Jonghyun Lee, **Taewoon Kang**, Pratheek B, Yongqin Wang, Rachit Rajat, Mengyuan Li, Gunjae Koo, Daniel Wong, Murali Annavaram. "FLASH: Fault-Led, Attested, Secure Hybrid-Key Checkpoint for Confidential GPU." *Proceedings of the 2026 ACM Annual Technical Conference*, 2026. [To appear]
5. **Taewoon Kang**, Namhun Kim, Jihun Lee, Gunjae Koo. "SoftmaxPIM: An HBM-Based PIM Architecture for Accelerating GeMV–Softmax Execution Pipeline." *IEEE Computer Architecture Letters*, 2026.
6. **Taewoon Kang**, Namhun Kim, Geonwoo Choi, Taeweon Suh, Gunjae Koo. "SparsePIM+: Accelerating SpMV on HBM-Based PIM via Logic-Die Accumulators with Opportunistic TSV Utilization." *Journal of Systems Architecture*, 2026.
7. Sunchae Kim, **Taewoon Kang**, Sangwon Shin, Taeweon Suh, Yibin Yang, Gunjae Koo. "Sumcheck-PIM: An Efficient HBM-Based PIM Architecture for Linear Complexity Zero Knowledge Proofs." *ACM International Conference on Supercomputing*, 2026.
8. **Taewoon Kang**, Geonwoo Choi, Taeweon Suh, Gunjae Koo. "SparsePIM: An Efficient HBM-Based PIM Architecture for Sparse Matrix-Vector Multiplications." *ICS '25: Proceedings of the 39th ACM International Conference on Supercomputing*, 2025.

Publications | Domestic

1. Seongpil Yang, **Taewoon Kang**. "Performance Analysis of a Per-Row Activation Counting Mechanism in DRAM." *The Institute of Electronics and Information Engineers, 2025 Summer Conference*, 2025.
2. Sang-un Jo, **Tae-woon Kang**, Yongwoo Kim. "A Design and Implementation of MP3 Player using ARM Cortex-M0 DesignStart." *The Korean Institute of Electrical Engineers Workshop*, Yeosu, Korea, Jul. 2022.
3. Sang-un Jo, **Tae-Woon Kang**, Yongwoo Kim. "A design and verification of ARM Cortex-M0 DesignStart MCU system using open source IP." *ICS 2022*, Jeonju, Korea, Apr. 2022.
4. **Taewoon Kang**, Yongwoo Kim. "An Implementation of CNN-based Real-Time Face Mask Detector using Jetson Nano." *JKAIS*, Vol. 23, 2022.

Patents | Domestic

Gunjae Koo, Taewoon Kang. "Apparatus and Method for Sparse Matrix-Vector Multiplication Based on Processing-In-Memory"

KR-Registration No. 10-3008421

Awards and Honors

All awards below: Korea University, Korea.

Support Program for Outstanding Graduate Students' International Joint Training — Jan. 2026
NRF (Brain Korea 21), Korea

All awards below: Sangmyung University, Korea.

The top prize of CISC Career day — May 2022

Header (in one's dept.) Scholarship — Mar. 2022

Header (in one's dept.) Scholarship — Sep. 2021

"BONGSA" Scholarship (Leader of Department) — Sep. 2021

Secondary Header (in one's dept.) Scholarship — Mar. 2021

Work Scholarship — Mar. 2021

"BONGSA" Scholarship (Leader of Department) — Mar. 2021

Secondary Header (in one's dept.) Scholarship — Sep. 2020

Coursework

- Parallel processing; AI accelerator architecture
- Advanced computer architecture; storage system architecture
- ARM Cortex-M series IP implementation
- Electrical circuit; microelectronic circuit
- Semi-custom ASIC design with Synopsys design tool; VLSI design
- Digital logic circuit; computer architecture and design; design with Verilog; SoC design
- Semiconductor physics; semiconductor fabrication process
- Calculus; engineering mathematics

Skills and Techniques

- C, C++, Python
- GEM5, DRAMsim3
- Chip design verification with Xilinx/Altera FPGA
- Python (PyTorch); C language (Atmel, Keil)
- Verilog 2001; SystemVerilog
- ARM Cortex-M0/M3 implementation
- RISC-V simple processor design
- Semi-custom ASIC design with Synopsys EDA tool

Side Projects

Developed and released **PDFListener** (pdflistener.me) for macOS, iPhone, and iPad using AI vibe coding. Reads English research PDFs aloud with local speech synthesis and sentence highlighting.